

INDIVIDUAL ASSIGNMENT 2

ML Pipeline Design & MLOps Analysis

Course	DDM501 - AI in DevOps, DataOps, MLOps
Assignment	Individual Assignment 2
Weight	5%
Submission	Written report

I. Overview

This assignment focuses on **design, analysis, and evaluation** of ML pipelines and MLOps practices. Building upon the system design from Assignment 1, it is applied concepts from Sessions 3-5 to design comprehensive pipeline architectures and critically analyze MLOps tools and metrics. Also, a code snippet will be included to demonstrate the concepts and design decisions.

Continuing from Assignment 1

You must:

- Reference the previous design document briefly (problem statement, architecture overview)
- You may refine your architecture based on new insights from Sessions 3-5

II. Report Requirements

- a. Your report must include the following sections.

Section	Requirements
Pipeline design (25%)	Design a comprehensive ML pipeline architecture: Pipeline architecture diagram - Create a detailed pipeline diagram showing all stages and data flow - Include: data ingestion, validation, preprocessing, feature engineering, training, evaluation, and deployment stages - Show connections between components and data transformations Stage specifications For each pipeline stage, describe the <i>purpose, input/output, key operations, quality gates</i>. Design rationale: - Justify your pipeline structure choices - Explain how components are modular and reusable - Discuss error handling and recovery strategies - Address scalability considerations
Experiment tracking & metrics analysis (25%)	Design an experiment tracking strategy and analyze how metrics guide model selection: Experiment design

	- Experiment variables: Algorithms, hyperparameters, features, data preprocessing - Baseline definition: baseline model - Experiment matrix: Create a table showing planned experiments (minimum 8-10 experiment configurations) Metrics strategy - Primary metrics: Metrics determine model success - Secondary metrics: Additional metrics provide insights - Business-model alignment: Model metrics connect to business goals Code snippets - Mlflow experiment setup - Parameter and metric logging: hyper-parameters, metrics and artifacts - Model registration: How to register a model and versioning Results analysis Create an experiment results table and analyze: - Generate results for your experiments - Analyze patterns: Which configurations perform best? - Discuss trade-offs between different metrics - Recommend the best model configuration with justification
Workflow orchestration design (20%)	Design the workflow orchestration for your ML pipeline: DAG Architecture - DAG Diagram: Visual DAG to show tasks - Task Descriptions: Describe each tasks - Dependencies: Explain tasks depend on each others Scheduling Strategy - Trigger type - Frequency Code snippets: - Airflow DAG definitions - Task definitions - Task dependencies Operational considerations: - Failure handling - Alerting - Monitoring
Code quality & documentation (20%)	Code standards: - Code style - Docstring - Type hints

	- Comments Configurations: - Config file example - Config loading - Environment variables
Reproducibility & Versioning Strategy (10%)	Design a strategy to ensure reproducibility: Versioning Strategy - Code Versioning: Git branching strategy, release management - Data Versioning: How to track data changes over time - Model Versioning: Model registry structure, naming conventions Code snippets: - Code versioning - Random seed settings - Dockerfile

III. Submission Guidelines

Submission Process

- **File naming:** DDM501_Assignment2_[StudentID]_[Name].pdf